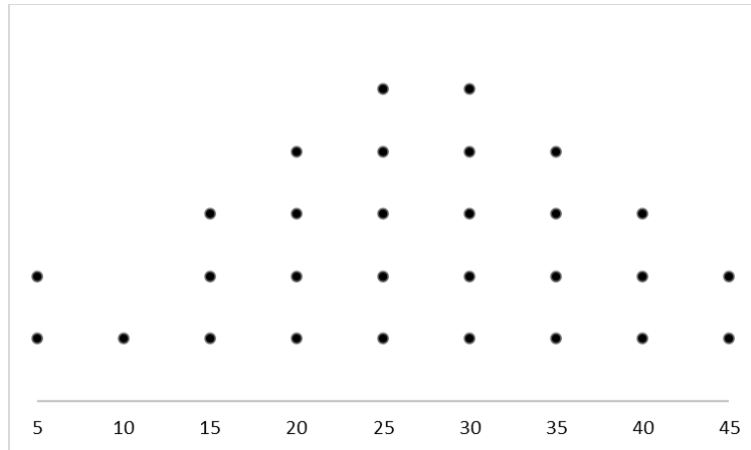


Ronaldo wanted to know how many minutes of exercise his friends did each week. His data is shown in the dot plot below.



Minutes

1. Explain which measure of center, the mean or the median, best describes the center of the data.

Because the data is somewhat skewed, the median best describes the center of the data.

2. Find the value of the measure you chose in question 1.

25

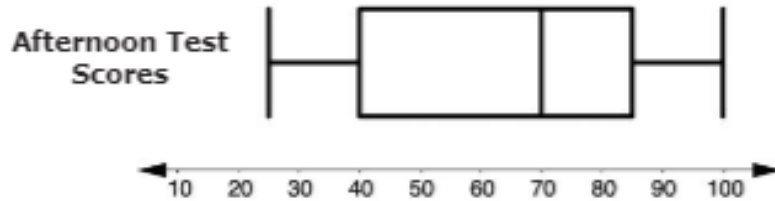
3. Describe the variation of the weekly exercise times.

The majority of Ronaldo's friends exercise from 20 – 35 minutes.
The total range is 40.

4. Does the IQR or the range, better represent the variation in exercise times? Explain your reasoning.

Because the data is somewhat skewed, the IQR better represents the variation in exercise times.

Mr. Arias gave his students a quiz in the afternoon. His students' scores are shown in the box plot shown below.



5. Which measure of center best summarizes the data?

Median because there seems to be some lower scores that are skewing the data.

6. Which measure of variation should Mr. Arias use to best summarize his data? Explain your reasoning.

IQR because there are some outliers that are skewing the data.

7. Mr. Arias determined taking a test in the afternoon was good for his students. Do you agree or disagree? Explain your reasoning.

Disagree because 1st quartile range is larger than the third which means quite a few students did not do very well.

Gia decided to record the amount of time it took her friends to get to school from their houses. The times are shown in minutes.

19, 36, 12, 23, 45, 2, 17, 20

8. What measure of center best describes the typical amount of time it takes Gia's friends to get to school? Explain your reasoning.

The median because it's not affected by the extreme values of 2 and 45.

9. By looking at the data, Gia thinks that range best summarizes the variation of the data. Do you agree with her?

No because the outliers will affect the range.

10. Which measure of variation best describes the typical amount of time it takes Gia's friends to get to school? Explain your reasoning.

IQR because it provides the variation of the center (majority) of data.